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CHAPTER 12

ASPHALT IN PAVEMENT MAINTENANCE

This chapter provides an overview of the methods used to maintain hot mix asphalt pavements. Pavement maintenance is the work performed to keep a pavement as close as possible to its as-built condition, under normal traffic and weather conditions.

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12.1 Introduction

This chapter provides practical information about methods and terminology that applies to the use of asphalt in pavement maintenance.

Large differences in soil types, climate, traffic, and other factors make for a variety of problems, even within small areas. Some regions have heavy rainfall, while others are hot and dry; some pavements must accommodate vehicles carrying heavy loads, while others are subjected to only lightweight traffic.

Despite these differences, there are maintenance methods that can be used equally well in all regions. This chapter provides an overview of the methods used to maintain hot mix asphalt (HMA) pavements.

There are about two million miles of paved roads across the United States. All require maintenance. The original investment in these structures was in the billions of dollars. Their maintenance now costs millions of dollars annually. Obviously, timely and proper maintenance is imperative. Pavement maintenance is the routine work performed to keep a pavement, under normal traffic conditions and forces of nature, as nearly as possible in its as-constructed condition. Preventive maintenance not only preserves the pavement surface but also prevents early deterioration. It should not be regarded as a temporary expedient but as an investment in the pavement structure and as insurance against costly pavement renovation.

The three categories of pavement maintenance operations are the following:

- **Preventive maintenance** is the routine work done to prevent deterioration of a pavement. It includes crack sealing, surface treatments, thin overlays, drainage maintenance, etc. The idea behind preventive maintenance is to expend funds and materials while the pavement is in relatively good condition in order to retard deterioration and thus prolong the life of the pavement. This can be done at a very reasonable cost. A good analogy for preventive maintenance is the painting of a house. When the paint fades, it is repainted; the responsible homeowner does not wait until the paint is cracked and peeling. As with painting a house, the effectiveness of the preventive maintenance is directly related to the condition of the pavement. Conducting preventive maintenance activities on a sound pavement in good condition is very effective. Conducting an inappropriate repair (in either method or timing) can accelerate the rate of pavement deterioration. In summary, preventive maintenance is completing the right maintenance activity at the right time.
- **Corrective maintenance** consists of those activities that are accomplished after a deficiency occurs. It is performed when the pavement needs repair, and it is the most costly kind of maintenance. It is

generally reactive and performed to correct a specific pavement distress. An example of this type of maintenance would be a chip seal to repair a low-friction condition.

- **Emergency maintenance** consists of those activities performed in an emergency situation. An example would be the repair of a severe pothole on a high-volume roadway.

Pavement maintenance is a portion of an agency's overall pavement management procedures. Pavement management is discussed in detail in Chapter 11, "Pavement Management."

12.2 Patching

Most pavements require patching at some time. If holes do not occur from traffic loading and natural causes, then service cuts and trenches produce them. The kind of defects requiring patching vary from cracked areas and shallow abrasions to deep holes.

Patching requires skill and close supervision. Prompt repair of small breaks helps to control expenditures because once an area is broken and water enters the subgrade, a large failure will result.

12.2.1 Materials

Hot Mix Asphalt

HMA mixes make long-lasting patches that rapidly develop stability. They should be used whenever it is practical and economical. See Chapter 4, "Hot Mix Asphalt (HMA) Design," and Chapter 7, "Manufacture of Hot Mix Asphalt."

Cold Mixes

Other maintenance materials are cold-plant asphalt mixtures employing emulsified asphalt for immediate use or for short-term stockpiling. Slow-curing cutback asphalt or emulsion containing solvent may be used for stockpiled patching mixes. When using cutback, be sure to check local environmental regulations. See Chapter 15, "Special Uses of Asphalt and HMA."

Proprietary Mixes

Producers have developed proprietary mixes, with special blends of aggregate and specialized binders, for specific purposes. As a rule, these mixes work well when used for their intended purpose. Information on these mixes can be found in trade magazine advertisements.

Specialty Mixes (Innovative State Highway Mixes)

Many state Department of Transportation (DOT) maintenance divisions have innovative mixes they have developed for pothole patching, spalled concrete joints, skin patching, etc. The DOTs are a good source of information about these specialty mixes.

In general, it is prudent to check with the local DOT to determine what materials are recommended, especially regarding proprietary mixes.

12.2.2 Full-Depth (Deep) Patching

Full-depth patching is the removal of the entire pavement surface layer, regardless of its thickness, over the area to be patched. Deep patching is

the removal of 4 inches (100 millimeters) or more of the pavement surface course (of the area to be patched) when the surface course is thicker than the planned patch. The procedures are the same whether for a full-depth or a deep patch. Full-depth patching procedures apply to either flexible or rigid pavements, but deep patching applies only to flexible pavements.

Full-depth (and deep) patches are used for making permanent pavement repairs. The material in the repair area should be removed to the depth necessary for reaching firm support as shown in Figure 12.1. This may mean removing some of the subgrade. It may also require installing additional drainage. The excavation should extend at least 1 foot (0.3 meter) into the good pavement surrounding the area to be patched. Patches should be square-edged and the cuts should be rectangular in shape. A pavement saw makes a fast and clean cut. If the asphalt layer to be removed is thicker than the depth of the sawcut, the remainder may be jack-hammered out. The vertical faces of the patch are then given a tack coat as shown in Figure 12.2.

For best results, the patch should be backfilled with a dense-graded HMA (see Figure 12.3). Then the mix should be spread carefully to prevent segregation (see Figure 12.4). If HMA is not available, an appropriate cold mix, specialty mix, or proprietary mix should be used. If the patch is more than 6 inches (150 millimeters) deep, the backfill should be placed in layers and each layer compacted thoroughly. Proper compaction is a major factor in producing a permanent patch. A vibratory-plate compactor is excellent for small patches. A roller may be more practical for large areas. The repair is then completed by placing the HMA surface layer and compacting it flush with the surrounding pavement surface. The patch should not be overfilled in anticipation

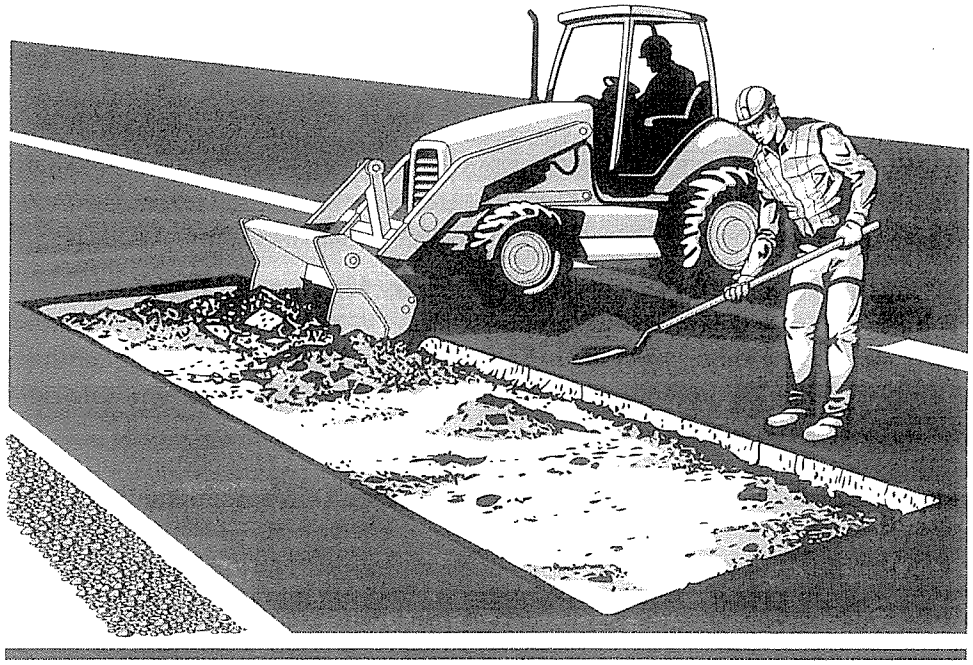


Figure 12.1 Removing the Surface and Base

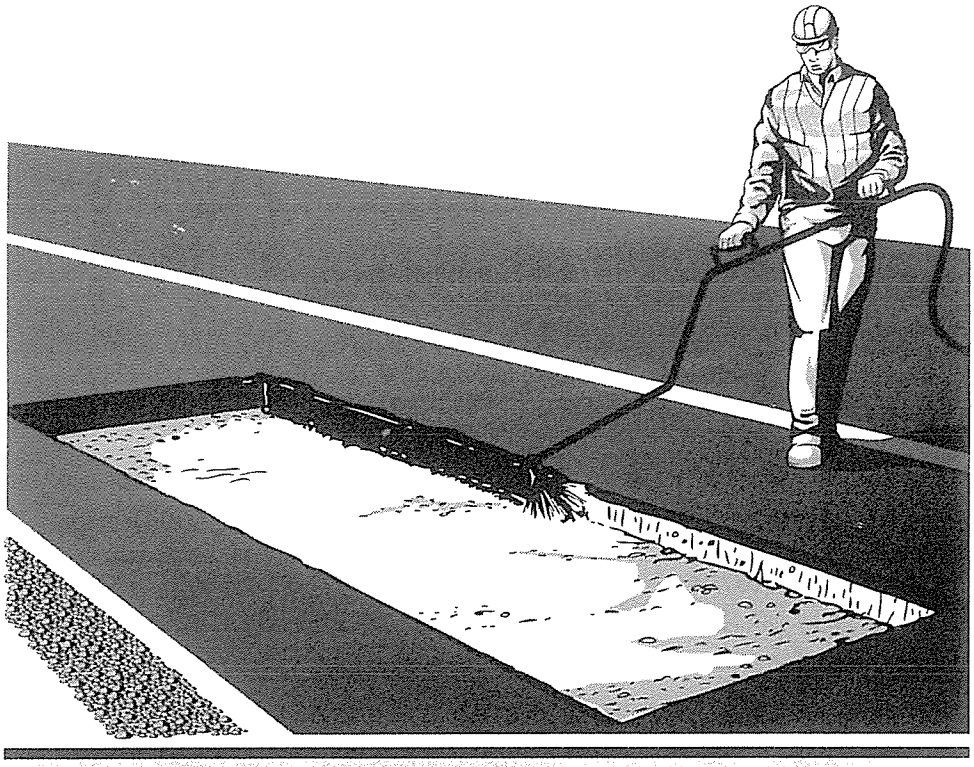


Figure 12.2 Applying a Tack Coat to the Vertical Sides

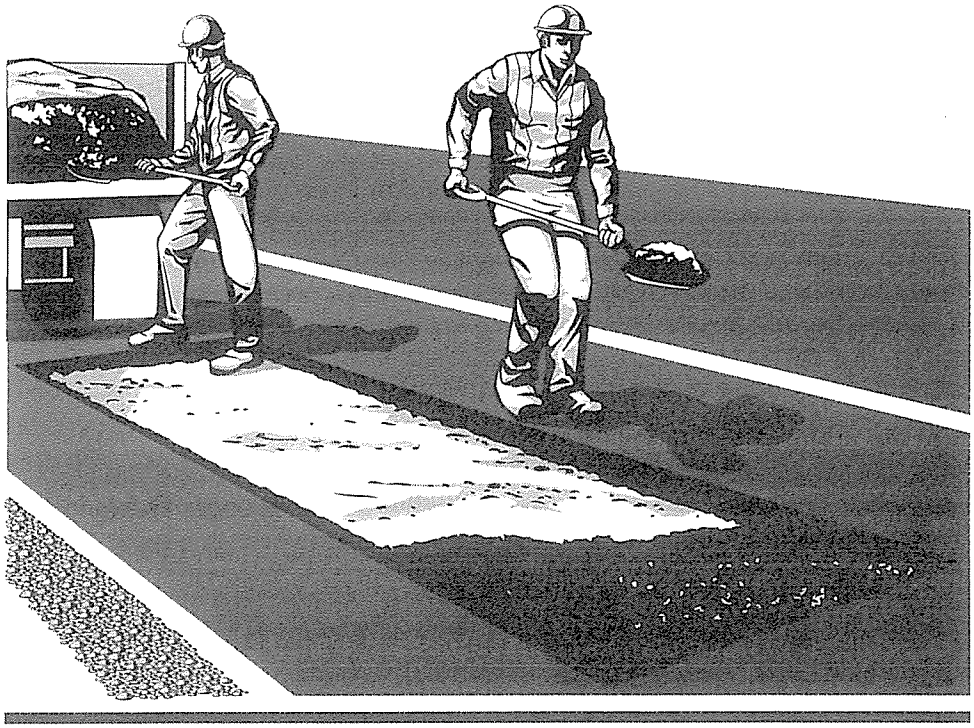


Figure 12.3 Backfilling the Patch Area with HMA



Figure 12.4 Spreading the Mix Over the Patch Area

of traffic compaction. A straightedge or string line should be used to check the evenness of the surface, as shown in Figure 12.5. If the patch is constructed using a cold mix, the surface of the patch may be given a sand seal coat.

A "patcher truck" is a useful piece of equipment for this type of maintenance work (see Figure 12.6). A patcher truck has bins and tanks for storage of HMA, liquid asphalt, and other materials. The bins and tanks are heated so the HMA and liquid asphalt are maintained at proper application temperatures. Tools such as a pneumatic hammer, a vibrating plate compactor, etc., are located on the truck, thus making it a self-contained patching unit.

Another piece of equipment that can greatly facilitate patching work is a cold-milling machine. When large and numerous patches are required, a medium-size milling machine (Figure 12.7) works well. When small and numerous patches are required, a small milling machine (Figure 12.8) does an excellent job.

12.2.3 Surface Patch (Skin Patch)

Surface patches are temporary repairs and should be used only on pavements that are in relatively good condition and have adequate thickness. The

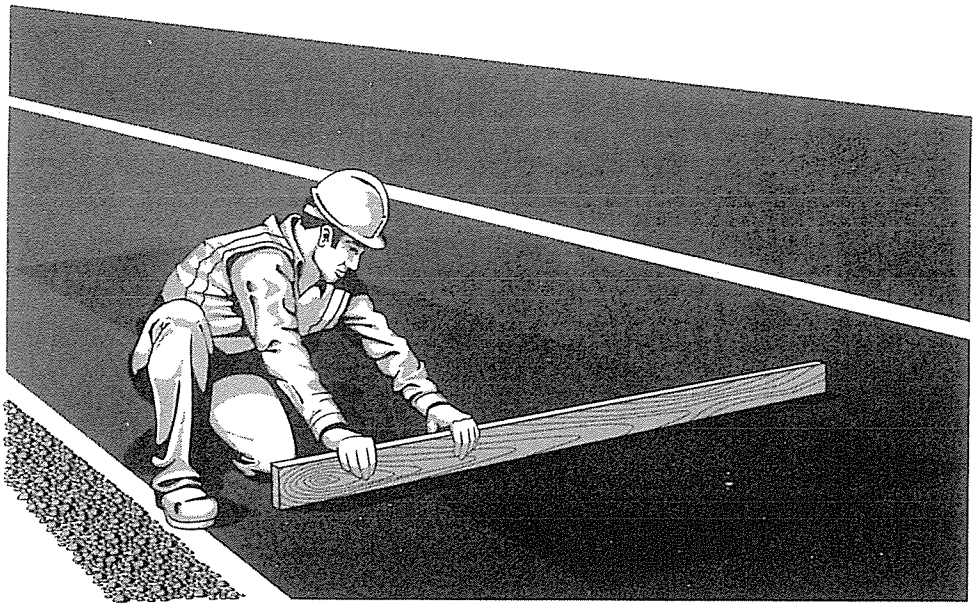


Figure 12.5 Checking Patched Surface with a Straightedge



Figure 12.6 Patcher Truck



Figure 12.7 Medium-Size Milling Machine



Figure 12.8 Small-Size Milling Machine

pavement should be at least 4 inches (100 millimeters) thick. A surface patch can be constructed without excavating the existing surface. The patch material should be feather-edged to match the in-place grade lines. When this procedure is used, it is a good idea to outline the patch area with a small trench produced by a pneumatic hammer. This will help tie the patch to the existing pavement and reduce the possibility of raveling.

Surface patches can also be constructed by milling a portion of the pavement (see Figure 12.9). This eliminates the need to feather-edge the patch to avoid raveling. It also provides additional depth, which will accommodate mixes with larger-size aggregates, thus producing a more stable patch. In all cases, the patch must match the elevation of the surrounding pavement.

The patching procedure consists of marking the area to be patched, making sure it extends 1 foot (0.3 meter) into good sound pavement.

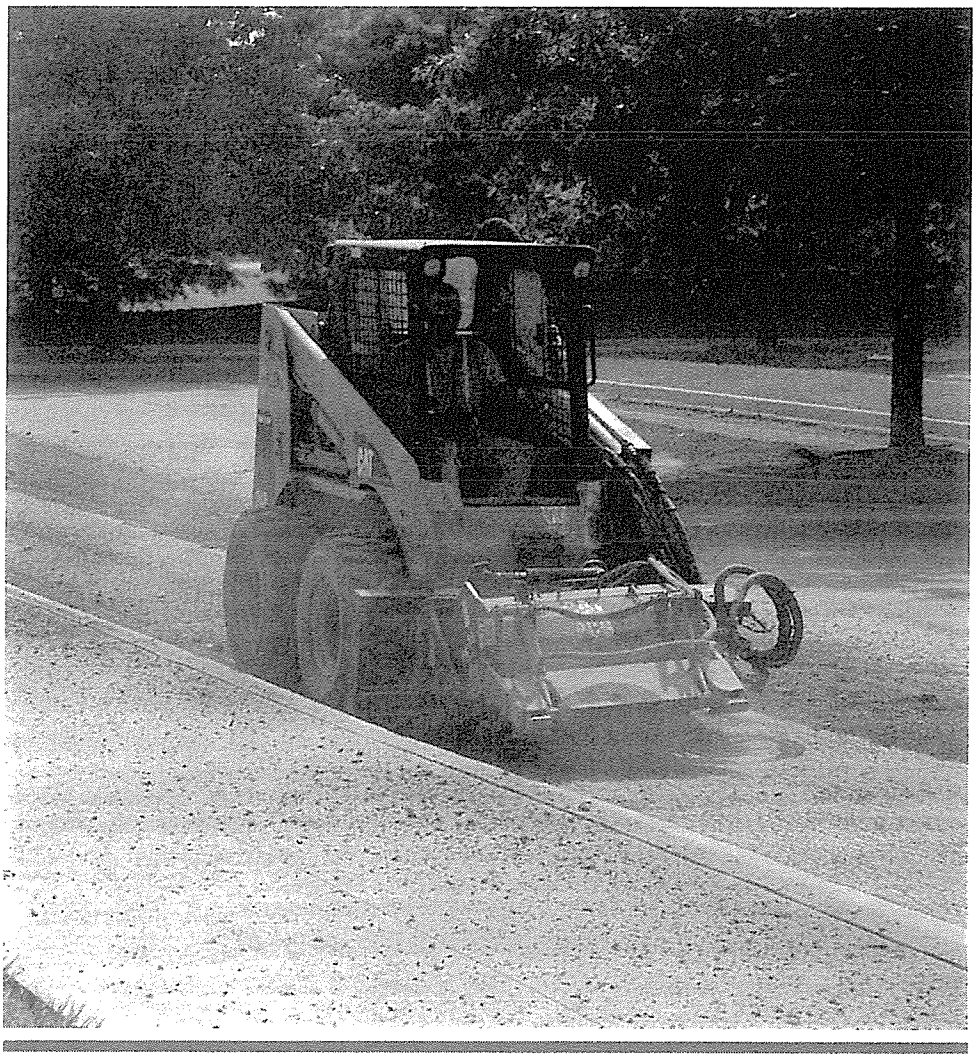


Figure 12.9 Skid Steer Loader Equipped with a Small Milling Head

If the procedure involves pavement removal, mill to a depth necessary to remove the distressed material. Thoroughly clean the area with high-pressure air, brooms, and/or water. After the area is clean and dry, apply a tack coat to the edges and the bottom of the patch area.

After the tack coat has set, rapidly place the HMA over the patch area. For surface patches, a fine-aggregate or sand mix is preferable, especially if the patch is placed without milling the existing pavement. Without milling, the surface patch must be feathered to zero thickness at the edge of the patch. Compact the patch with equipment appropriate for the size of the patch. When compacted, the patch should be level with the surrounding surface. A fog seal should be applied to the feather-edges to reduce the possibility of raveling.

12.2.4 Spray-Injection Patching

Spray-injection patching has existed since the early 1980s. The procedure gained prominence after the Strategic Highway Research Project of the late 1980s and early 1990s determined that it is a viable method of repairing small pavement defects with semipermanent repairs, particularly during inclement weather (wet and/or cold weather).

This procedure requires a specially equipped truck or trailer mounted unit (see Figure 12.10).

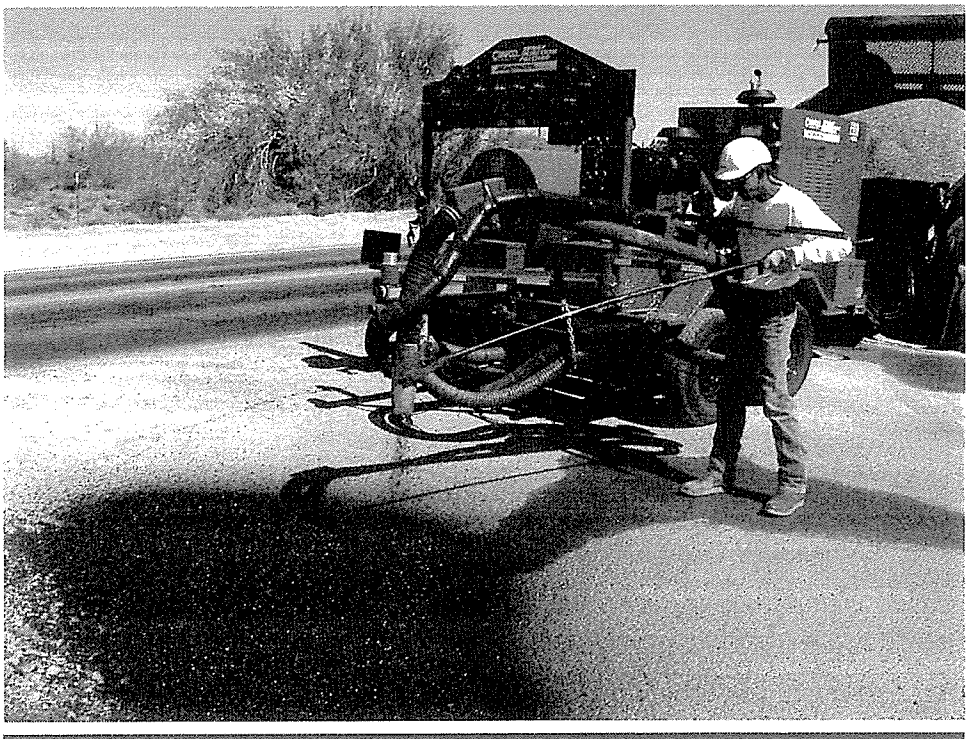


Figure 12.10 Spray-Injection Patching Machine

This unit consists of an emulsion tank, aggregate tank, heating components, high-volume blower, telescoping boom with injection head, and the necessary controls.

The procedure is essentially a one-person operation. It consists of cleaning the area to be patched with compressed air to remove loose material and debris, applying a tack coat of hot asphalt emulsion, blowing the combined aggregate and hot emulsion into the patch with forced air, and treating the top of the patch with a dust coat of aggregate.

The aggregate used in this procedure is essentially a one-size aggregate similar to a chip-seal aggregate. Compaction of the mix is accomplished by the force of the air when the mix is sprayed into the patch. This procedure is especially adaptable to pothole patching. The success or failure of this process depends largely on the operator's skill level.

12.3 Crack Sealing/Filling

Crack sealing/filling has two purposes:

- To prevent the intrusion of water into the underlying layers of the pavement structure, thus further weakening the pavement
- To prevent incompressible materials from entering the crack, thus causing further deterioration as the pavement expands and contracts with temperature changes

Crack sealing is usually done with liquid asphalt, cutbacks, or asphalt emulsions, and is considered a temporary procedure. These crack sealers usually maintain a tightly sealed crack only as long as the pavement remains stationary, both longitudinally and vertically. Once expansion and contraction or excessive vertical movement occurs, the seal's effectiveness can be lost and the cracks must be resealed. In climates with severe temperature changes, resealing may be necessary every year. In moderate or mild climates, resealing will be necessary less often. Although the cost per unit of this type of crack sealing is relatively low, the frequency with which it must be done might dictate a more lasting solution.

Crack filling is accomplished using a specially prepared hot poured sealant. Crack filling is considered a permanent seal. Three to eight years is the expected life, depending on the material used, the pavement condition, and the techniques used when applying the crack seal.

Along with proper drainage, crack sealing/filling is probably the single most important maintenance activity. Many pavement distresses can be traced to the intrusion of water into the pavement structure. Keeping water out of the pavement is essential to delay or prevent more serious distresses.

Some guidelines regarding crack sealing follow, but the details may vary according to local policies:

- **Hairline cracks** are ⅛ inch (3 millimeters) or less in width and too small to seal effectively. If numerous hairline cracks cover an area, a surface seal can be used for the repair. Fog seals, chip seals, slurry seals, and sand seals work well. This procedure is sometimes referred to as mass crack filling. The surface treatment selected must be fluid enough to flow into the hairline cracks.
- **Small cracks** are ⅛ inch (3 millimeters) to ½ inch (12.5 millimeters) in width and usually routed to a width of ¼ inch (6 millimeters) or greater to provide a reservoir for the sealant. The crack is then cleaned and sealed. If the cracks are more than 2 inches (50 millimeters) deep, a backer rod should be installed to conserve sealant.
- **Medium cracks** are ½ inch (12.5 millimeters) to ¾ inch (19 millimeters) wide and usually require only cleaning and sealing. A backer rod should be installed in cracks that are more than 2 inches (50 milli-meters) deep.

- **Large cracks** are wider than ¾ inch (19 millimeters) and should be filled with an asphalt emulsion slurry seal material, an HMA sand mix, or a hot poured sealant.

12.3.1 Materials

The following materials can be used for crack sealing:

- Liquid asphalt—cutback asphalt; MC-30, 70 and 250; emulsified asphalt; SS-1, SS-1h, CSS-1, CSS-1h, MFMS-1, and MFMS-1h
- Modified-asphalt crack sealer, ASTM D 6690, or equal
- HMA mixtures; slurry seal and hot sand mixes
- Blotter sand

12.3.2 Crack-Sealing Procedures

Clean the cracks using high-pressure air, sandblasting, wire brushing, hot-air blasting, or high-pressure water.

Cleaning is an essential step, because when the crack is not thoroughly dried and cleaned, the sealant will not adhere to the sides of the crack. The most effective procedure is sandblasting, but it is very labor intensive. Hot-air blasting is done with a hot compressed-air lance or heat lance connected to an air compressor (see Figure 12.11). This method helps dry the crack, and when the sealing operation follows closely behind the hot-air drying, the heated crack surfaces help the sealant adhere to the crack. Care should be taken when using a heat lance. The heat lance produces superheated air that can burn the binder in the mix along the crack surfaces. This will cause early failure of the sealant.



Figure 12.11 Cleaning of Cracks with Hot Compressed Air

After cleaning, the cracks should be checked for depth. A backer rod should be placed in large deep cracks to conserve sealant. The backer rod should be a compressible, non-shrinking, nonabsorbent material with a melting point higher than the temperature of the sealant. The backer rod should be about 25 percent wider than the crack so that it does not slip into or float out of the crack after installing the sealant.

Once the cracks are prepared, they are sealed with liquid asphalt. The material may be applied using hand-carried pour pots, wheel-mounted pour pots, or the wand of a small distributor. Pour only the amount of material that will seal the crack. Do not try to completely fill the crack because this is a waste of sealant. The idea is to just coat the surfaces of the crack, with a small amount of sealant settling in the bottom of the crack (see Figure 12.12).

Any spillage or excess material on the pavement surface should be removed with a squeegee. Any surface so cleared of material should also be dusted with blotter sand to prevent tracking of the sealant by traffic.

This procedure applies to small, medium, and large cracks. Hairline cracks will not accept sealant, and because hairline cracks are so narrow, significant amounts of moisture will not penetrate anyway. If hairline cracks are numerous, a mass crack-filling procedure should be used.

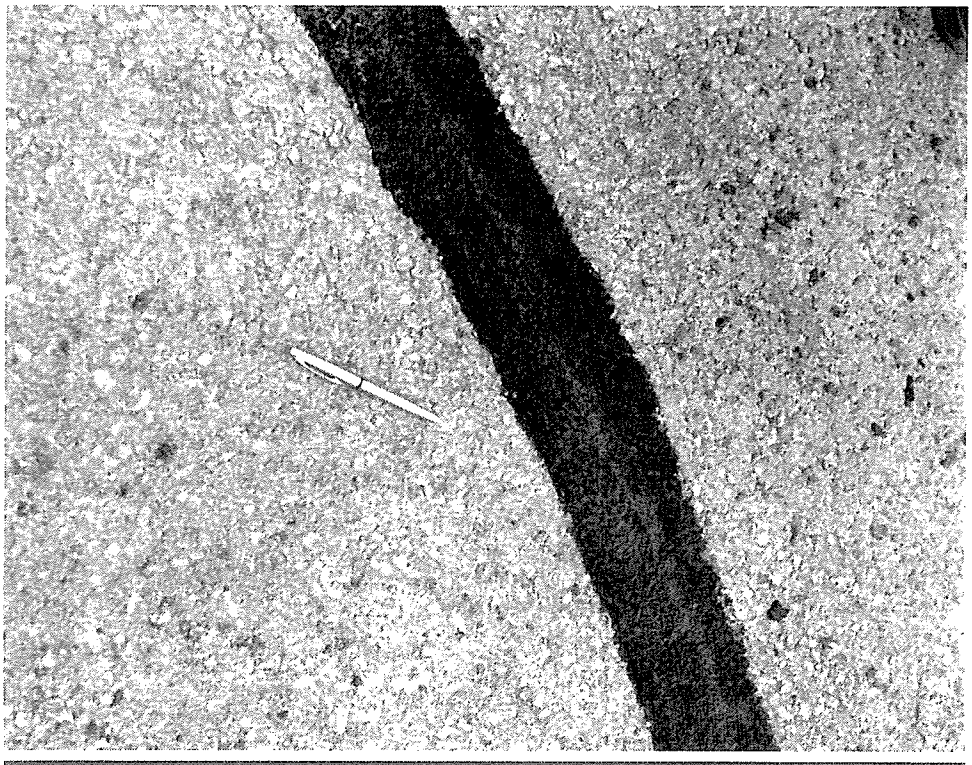


Figure 12.12 Properly Sealed Crack

12.3.3 Crack-Filling Procedures

In crack-filling procedures, all cracks, whether hairline, small, medium, or large, are filled. First, the cracks need to be routed out to the sealant manufacturer's width-to-depth ratio specifications.

All debris from the routing operations must be cleared as previously described. Backer rods should be placed in deep cracks to lessen the quantity of relatively expensive sealant needed. These backer rods must be resistant to the high temperatures of the hot-poured sealers.

Equipment used for crack sealing varies from truck- or trailer-mounted pressure applicators to hand wands (see Figure 12.13). The equipment is capable of heating and maintaining the temperature of the sealant at a relatively high temperature—the maximum being about 450°F.

The sealant should be applied from the bottom to the top of the crack to prevent air bubbles from forming and creating a weak spot in the sealant. To prevent tracking, the sealant should be ⅛–¼ inch (3–6 millimeters) below the top of the crack. Use a squeegee to remove excess sealant on the pavement surface, and then blot.



Figure 12.13 Placing Crack Filler with Hand Wand

12.4 Asphalt Surface Treatments

Asphalt surface treatments are versatile, widely used forms of asphalt construction and maintenance. As the following sections indicate, there are several forms of treatments and seal coats. When properly designed, they are economical, easy to place, and long-lasting. They seal and add life to road surfaces, but each type has one or more special purposes. A surface treatment is not a pavement. Rather, it provides a protective, waterproof cover over the underlying structure, which helps to resist traffic abrasion. It adds little or no load-carrying strength and, therefore, is not usually taken into account in computing the load-carrying capacity of a pavement. While a surface treatment can provide an excellent surface if it is used for the correct purpose, it is not a solution for all pavement problems. A clear understanding of the advantages and limitations of asphalt surface treatments is essential for best results. It is vital that a careful study of traffic requirements, along with an evaluation of the condition of the existing pavement, be made before deciding whether a surface treatment should be used.

Asphalt surface treatment is a broad term embracing several types of asphalt applications with or without a cover of mineral aggregate. Surface treatments are typically less than 1 inch (25 millimeters) thick. Asphalt binders used in these surface treatments are discussed in "Emulsified and Cutback Asphalts" in Chapter 2.

Surface treatments provide a waterproof cover over the existing pavement surface and provide resistance to abrasion by traffic. Specific functions of surface treatments are the following:

- To provide long-lasting, economical surfaces for granular-base roads having light and medium traffic volumes
- To prevent surface water from penetrating granular bases or old pavements that have become weathered or cracked
- To plug voids, and to coat and bond loose aggregate particles in pavement surfaces
- To renew a surface and restore skid resistance to traffic-worn pavements in which the surface aggregate has become polished
- To restore weathered surfaces and give new life to oxidized pavement surfaces
- To provide temporary cover in cases of delayed pavement construction or planned stage construction
- To control dust on low-volume roads
- To promote adherence of subsequent asphalt courses to granular bases (prime coats)

- To ensure a bond between subsequent pavement layers being placed or overlays of existing surfaces (tack coats)
- To improve the aesthetics of an existing pavement

The treatments discussed in this chapter include the following:

- **Spray applied seal** is an application of a diluted asphalt emulsion without a cover aggregate. It is used to seal and enrich the asphalt pavement surface and seal minor cracking.
- **Chip seal** is an application of asphalt followed immediately with an aggregate cover. Applications of two layers are referred to as a double chip seal and applications of three layers are referred to as a triple chip seal.
- **Slurry seal** is a mixture of fine aggregate, asphalt emulsion, water, and mineral filler (generally Portland cement).
- **Microsurfacing** is sometimes referred to as a polymer-modified slurry seal. The asphalt in a slurry seal is cured through a thermal process, but in microsurfacing, the curing process is chemically controlled.

12.4.1 Pavement Surface Preparation

Before any work is started, a thorough surface examination is made to determine needed repairs and to evaluate requirements for treatment. Soft spots that may result in future failures, as well as variations in width, cross section, and profile that would leave the road unsightly, are noted. Side drainage defects are especially critical. If the surface being reconditioned is old, it should be examined for potholes, cracked areas, depressions, slick or bleeding asphalt areas, absorbent areas, and other surface defects.

Repairs and corrections must be made so the old surface will be in sound condition before treatment is begun. Some old pavements do not need repairs before they are resurfaced. Others may require complete scarifying, mixing, and recompacting, along with the removal of distressed material. However, most pavements need some patching and removal of excess asphalt.

When surface repairs such as patching are performed, sufficient time should be allowed before treating the surface to ensure consolidation under traffic.

The asphalt may not adhere to the pavement, unless the surface to be covered is completely clean. It is therefore necessary to clean the surface before spraying the asphalt emulsion. A power street sweeper (Figure 12.14) is recommended to pick up both dust and loose particles, but if one is not available, a rotary power broom can be used. When brooms are used, flushing with water may be necessary to meet clean air standards or to remove caked-on materials.

Power sweepers or brooms are also used to remove loose particles after the treatment is completed and the asphalt has properly cured. Only light broom pressure should be used so that the aggregate particles embedded in the asphalt membrane are not dislodged.



Figure 12.14 Preparation of Surface with Power Sweeper

12.4.2 Spray-Applied Sealers

Fog Seals

A fog seal is a light application to an existing surface of a slow-setting asphalt emulsion diluted with water. It can be diluted in varying proportions—up to one part emulsion to five parts water, but in most cases, a one-to-one dilution is used. Grades of asphalt emulsion normally used for this purpose are SS-1, SS-1h, CSS-1, or CSS-1h.

A fog seal can be a valuable maintenance aid when used for its intended purpose. It is not a substitute for an asphalt-aggregate surface treatment or a slurry seal. It is used to renew old HMA pavement surfaces that have become dry and brittle with age, to seal small cracks and surface voids, and to inhibit raveling.

The fairly low-viscosity diluted emulsion flows easily into the cracks and surface voids. It also coats aggregate particles on the surface.

This corrective action prolongs pavement life and may delay the need for major maintenance or rehabilitation.

The total quantity of fog seal used is approximately 0.10–0.15 gallon/square yards (0.45–0.70 liter/square meters) of diluted material. Exact quantities are determined by the surface texture, dryness, and degree of cracking or raveling of the pavement on which the fog seal is to be sprayed. Over application must be avoided because it will result in asphalt pickup by vehicles and possibly creating a slippery surface. If excessive emulsion is applied, a light dusting of the affected area with fine sand may remedy the problem.

Traffic must be kept off the fog seal until the emulsion breaks and is substantially absorbed into the existing surface. This curing period may range from one hour in hot, dry conditions to as much as three hours or longer in cool, humid conditions.

Rolling with a pneumatic-tire roller may reduce the curing period.

Rejuvenators

Asphalt consists of two fractions: asphaltenes, the solid fraction, and maltenes, the liquid fraction. During the aging process, the maltene oxidizes and the ratio of maltenes to asphaltenes is reduced. The aging process causes asphalt mixes to become dry and brittle. Asphalt rejuvenators are emulsions of these maltenes. Depending on the specifications (ASTM D 4552), the emulsion may contain only maltenes or maltenes with various amounts of asphaltenes.

Like fog seals, rejuvenators are used for preventive maintenance. The application rate of a rejuvenator is crucial. Trial tests must be done to determine the amount of rejuvenator the pavement will sufficiently absorb. Once the application rate is determined, care must be taken to assure that the correct rate is applied to the pavement.

Rejuvenators are also used as recycling agents to restore the aged binder in recycle mixes.

Dust Palliatives

A dust palliative is the application of a diluted asphalt emulsion directly on an unpaved surface as a dust control agent or palliative. SS-1, SS1h, CSS-1, or CSS-1h diluted with five or more parts of water by volume is used. The diluted material is sprayed with an asphalt distributor in repeated small applications as required. The application rates are approximately 0.1–0.5 gallon/square yard (0.45–2.3 liters/square meter). The total amount applied depends on the condition of the existing surface.

Some penetration of the emulsion is expected, and with a more open surface more emulsion can be applied. As with other spray treatments without cover material, small test sections are recommended to determine the best application rate for the existing conditions.

12.4.3 Chip Seals

A chip seal (sometimes referred to as a bituminous surface treatment) is used to protect the pavement from the deteriorating effects of sun and water. A secondary benefit of a chip seal is an increase in the skid resistance of the pavement surface. This occurs because the cover aggregate increases the surface texture of the pavement.

A single bituminous surface treatment is used as a wearing and waterproofing course. It consists of a sprayed application of asphalt immediately covered by a single layer of uniform-size aggregate. The thickness of the treatment approximates the nominal maximum-size aggregate particles used in the particular surface treatment.

When a spreader drops a one-size cover aggregate on an asphalt film, the particles will lay in an unarranged position. After compaction, the particles will become oriented into their densest position with about 20 percent voids between the particles. It is desirable to fill these voids two-thirds to three-fourths full with asphalt. A typical design will call for 70 percent of the voids to be filled. As the water evaporates from the asphalt emulsion, the residual asphalt will fill 55–60 percent of the voids between the aggregates under average conditions.

A multiple-surface treatment provides a wearing and waterproofing course of greater durability than a single-surface treatment. It consists of two or more alternate applications of asphalt and aggregate.

For each succeeding course, the nominal top size of cover aggregate should be not more than one-half the size of that for the previously placed course. No allowance is made for wastage. Also, after the first course, no correction is made for underlying surface texture.

Asphalt quantities for each course are added together; 40 percent of the total should be applied for the first application and 60 percent for the second application in a double-surface treatment. In a triple-surface treatment, 30 percent of the total should be applied for the first application, 40 percent for the second application, and 30 percent for the third application. In multiple-surface treatments, the first course of cover aggregate generally determines the thickness. Subsequent courses partially fill the upper voids in the previously placed courses.

Materials

Asphalt Binders

The selection of the proper type and grade of asphalt for a chip seal must include the following considerations:

- Temperature of the surface to which the treatment will be applied
- Air temperature
- Humidity and wind
- Condition of the surface
- Type and condition of the aggregate to be applied

The correct type and grade of asphalt for the chip seal will

- be fluid enough to spray and cover the surface uniformly, yet viscous enough to remain in a uniform layer and not puddle in depressions or run off the crown at grade;
- after application, retain the required consistency to wet the applied aggregate;
- cure and develop adhesion quickly;
- after rolling and curing, hold the aggregate tightly to the road surface to prevent dislodging by traffic; and
- when applied in the proper amount, not bleed or strip with changing weather conditions.

Certain grades of asphalt cements and emulsified asphalts satisfy these requirements. In hot, dry weather, soft grades of asphalt cement may be used

if dry aggregate is placed immediately after spraying. However, rapid-setting emulsified asphalts are usually the most suitable for surface treatments.

The types and grades of asphalts most suitable for single- and multiple-surface treatments are RS-1 and RS-2; CRS-1 and CRS-2; MS-1 and HFMS-1; PG 46–34; PG 52–28; PG 58–28; PG 64–22; and PG 70–22. Many asphalt suppliers are providing polymer-modified asphalt binders and emulsions for the construction of surface treatments.

Note: When PG asphalts are considered for use as the binder for a surface treatment, the use of heated pre-coat aggregates for the cover aggregate should be considered.

Aggregates

Most aggregates, such as gravel, crushed stone, and crushed slag, can be successfully used for cover. The aggregate selected, however, must meet certain requirements of size, shape, cleanliness, and surface properties.

Aggregate should be as uniform as economically practicable (a single size) so that the surface treatment will have essentially only one layer of aggregate. Generally, the largest size should be no more than twice the smallest size, with a reasonable tolerance in dimensions to allow for economical production. Aggregates having a maximum size of less than ½ inch (13 millimeters) in diameter provide a quieter and smoother riding surface than larger ones.

The ideal aggregate shape is cubical. A large number of flat or elongated particles is undesirable because they may be completely covered by asphalt applied in a quantity sufficient to hold the cubical particles.

It is important that the aggregate be clean. If the particles are coated with dust, silt, or clay, the coating forms a film that prevents asphalt-aggregate adhesion. Good adhesion between aggregate and asphalt, and retention of this adhesion are essential to a surface treatment's success. Adhesion, however, is influenced by many variables. Asphalt cement does not adhere well to wet aggregate. Best adhesion is obtained when aggregates are hot and dry. Some moisture may be advantageous when asphalt emulsion is used.

Surface Treatment Design

There are several theoretical procedures for determining the quantity of asphalt binder and cover aggregate for the appropriate surface treatment. These usually involve determining the average least dimension, the voids in the cover aggregate, and the bulk specific gravity. Mathematical calculations, coupled with laboratory testing, are typically used to determine the required quantities of asphalt and aggregate. The design procedures are available from ASTM and AASHTO; many state DOTs have their own procedures. The goal is to have aggregate particles to be approximately 70 percent embedded in asphalt. Adjustments must be made to account for the traffic volume, the absorption of the asphalt binder in the existing pavement and the cover aggregate, the texture of the existing pavement, and the characteristics of the cover aggregate (size, shape, and gradation).